

Chapter 8. Nested Designs and Analysis

- 8.1 Two-Way Nested Designs
- 8.2 Crossed Nested Designs

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8.1 Two-Way Nested Designs

- A large manufacturing company operates three regional training schools for mechanics. The schools have two instructors each who teach classes of about 15 mechanics in three-week sessions.
- The company was concerned about the effect of school (factor A) and instructor (factor B) on the learning achieved

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□ Factors:

- Three regional training schools
- Two teachers in each school
- 6 treatments (3×2)

- Experimental Unit: Twelve classes of about 15 mechanics in each
- Classes in each district were formed in the usual way and then randomly assigned to one of the two instructors in the school

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Factor A (school) <i>i</i>	Factor B (instructor) <i>j</i>		Total
	1	2	
Atlanta	25 29	14 11	
Total	$Y_{11} = 54$	$Y_{12} = 25$	$Y_{1..} = 79$
Chicago	11 6	22 18	
Total	$Y_{21} = 17$	$Y_{22} = 40$	$Y_{2..} = 57$
San Francisco	17 20	5 2	
Total	$Y_{31} = 37$	$Y_{32} = 7$	$Y_{3..} = 44$
	Total		$Y_{...} = 180$

?: How many teachers in the experiment

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(a) Crossed Factors

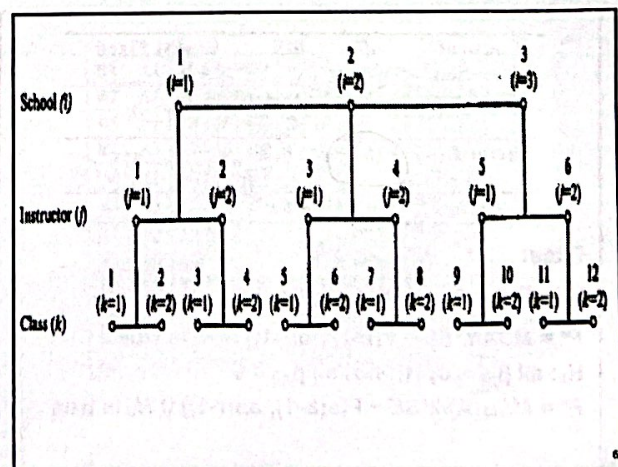
School (factor A)	Instructor (factor B)					
	1	2	3	4	5	6
Atlanta						
Chicago						
San Francisco						

(b) Nested Factors

School (factor A)	Instructor (factor B)					
	1	2	3	4	5	6
Atlanta						
Chicago						
San Francisco						

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嵌套设计



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ANOVA Model for two-way nested design:

Factor A	Factor B in Factor A					
A ₁	B ₁₍₁₎	B ₂₍₁₎	...	B _{j(1)}	...	B _{b(1)}
A ₂	B ₁₍₂₎	B ₂₍₂₎	...	B _{j(2)}	...	B _{b(2)}
⋮	⋮	⋮	⋮	⋮	⋮	⋮
A _i	B _{1(i)}	B _{2(i)}	...	B _{j(i)}	...	B _{b(i)}
⋮	⋮	⋮	⋮	⋮	⋮	⋮
A _a	B _{1(a)}	B _{2(a)}	...	B _{j(a)}	...	B _{b(a)}

$$Y_{ijk} = \mu + \alpha_i + \beta_{j(i)} + \varepsilon_{ijk}$$

$i = 1, 2, \dots, a; j = 1, 2, \dots, b; k = 1, 2, \dots, r;$

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Partition of total sum of squares and degrees of freedom

$$\sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{...})^2 = br \sum_i (\bar{Y}_{i..} - \bar{Y}_{...})^2 + r \sum_i \sum_j (\bar{Y}_{ij.} - \bar{Y}_{i..})^2 + \sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{ij.})^2$$

$$SSTO = SSA + SSB(A) + SSE$$

$$df_{TO} = df_A + df_{B(A)} + df_\varepsilon$$

$$abr - 1 = (a - 1) + a(b - 1) + ab(r - 1)$$

$$\begin{aligned} & (b-1) + (a-1)(b-1) \\ &= b-1 + ab - a - b + 1 \\ &= ab - a = a(b-1) \end{aligned}$$

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ANOVA table for two-way nested design

Source	Degree of Freedom	E(MS) Fixed	E(MS) Random	E(MS) Mixed
A	$a - 1$	$\sigma_\varepsilon^2 + \frac{br}{a-1} \sum_i \alpha_i^2$	$\sigma_\varepsilon^2 + r\sigma_\beta^2 + br\sigma_\alpha^2$	$\sigma_\varepsilon^2 + r\sigma_\beta^2 + \frac{br}{a-1} \sum_i \alpha_i^2$
B(A)	$a(b-1)$	$\sigma_\varepsilon^2 + \frac{r}{a(b-1)} \sum_i \sum_j \beta_{j(i)}^2$	$\sigma_\varepsilon^2 + r\sigma_\beta^2$	$\sigma_\varepsilon^2 + r\sigma_\beta^2$
Error	$ab(r-1)$	σ_ε^2	σ_ε^2	σ_ε^2

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Fixed Model

$$Y_{ijk} = \mu + \alpha_i + \beta_{j(i)} + \varepsilon_{ijk}$$

$$\sim N(\mu + \alpha_i + \beta_{j(i)}, \sigma_\varepsilon^2)$$

$i = 1, 2, \dots, a; j = 1, 2, \dots, b; k = 1, 2, \dots, r$

$$\varepsilon_{ijk} \sim N(0, \sigma_\varepsilon^2)$$

$$\sum_i \alpha_i = \sum_j \beta_{j(i)} = 0$$

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Cause of Variation	df	MS	E(MS) Fixed
Factor A	$a - 1$	MSA	$\sigma_\varepsilon^2 + \frac{br}{a-1} \sum_i \alpha_i^2$
Factor B(A)	$a(b-1)$	$MSB(A)$	$\sigma_\varepsilon^2 + \frac{r}{a(b-1)} \sum_i \sum_j \beta_{j(i)}^2$
Error	$b(r-1)$	MSE	σ_ε^2

F test

$H_0: \text{all } \alpha_i = 0; H_1: \text{not all } \alpha_i = 0$

$F^* = MSA/MSE \sim F(a-1, ab(r-1))$ if H_0 is true

$H_0: \text{all } \beta_{j(i)} = 0; H_1: \text{not all } \beta_{j(i)} = 0$

$F^* = MSB(A)/MSE \sim F(a(b-1), ab(r-1))$ if H_0 is true

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Data Training;

Input School \$ Instructor \$ Score @@;

Datelines;

A1B125 A1 B2 14
A1B129 A1 B2 11
A2B111 A2 B2 22
A2B16 A2 B2 18
A3B117 A3 B2 5
A3B120 A3 B2 2

PROC GLM;

Class School Instructor;

MODEL Score = School Instructor(School);

Contrast 'Instructor in A1' Instructor(School) 1 -1 0 0 0 0;

Contrast 'Instructor in A2' Instructor(School) 0 0 1 -1 0 0;

Contrast 'Instructor in A3' Instructor(School) 0 0 0 0 1 -1;

Means School/Tukey;

LSMeans Instructor(School)/StdErr PDiff;

RUN;

Instructor 交互作用 School 中

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Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	724.0000000	144.8000000	20.69	0.0010
School	2	156.5000000	78.2500000	11.18	0.0095
Instructor(School)	3	567.5000000	189.1666667	27.02	0.0007
Instructor (A1)	1	210.2500000	210.2500000	30.04	0.0015
Instructor (A2)	1	132.2500000	132.2500000	18.89	0.0048
Instructor (A3)	1	225.0000000	225.0000000	32.14	0.0013
Error	6	42.0000000	7.0000000		

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A1 而教 3 次 说明 明显 优于 A3.

Tukey Grouping Mean N School					
A	19.750	4	A1		
B A	14.250	4	A2		
B	11.000	4	A3		

Instructor	School	LSMEAN	SE	Pr> t	Number
B1	A1	27.0000000	1.8708287	<.0001	1
B2	A1	12.5000000	1.8708287	0.0005	2
B1	A2	8.5000000	1.8708287	0.0039	3
B2	A2	20.0000000	1.8708287	<.0001	4
B1	A3	18.5000000	1.8708287	<.0001	5
B2	A3	3.5000000	1.8708287	0.1106	6

I/J	1	2	3	4	5	6
1		0.0015	0.0004	0.0382	0.0183	0.0001
2	0.0015		0.1813	0.0298	0.0639	0.0145
3	0.0004	0.1813		0.0048	0.0092	0.1077
4	0.0382	0.0298	0.0048		0.5913	0.0008
5	0.0183	0.0639	0.0092	0.5913		0.0013
6	0.0001	0.0145	0.1077	0.0008	0.0013	

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Case Study (2)

□ Testing plant acid levels:

□ Factors:

■ A: plant (A1 A2 A3 A4)

■ B: leaf (B1 B2 B3)

■ 12 treatments (4 × 3)

■ 3 replications

□ Experimental Unit: 12 × 3 leaves

□ Response: Acid level

Plant i:	1			2			3			4		
Leaf j:	1	2	3	1	2	3	1	2	3	1	2	3
k=1:	11.2	16.5	18.3	14.1	19.0	11.9	15.3	19.5	16.5	7.3	8.9	11.3
k=2:	11.6	16.8	18.7	13.8	18.5	12.4	15.9	20.1	17.2	7.8	9.4	10.9
k=3:	12.0	16.1	19.0	14.2	18.2	12.0	16.0	19.3	16.9	7.0	9.3	10.5

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replication

Data Leaf;

Input Plant \$ Leaf \$ Acid @@;

Datalines;

A1 B1 11.2 A2 B1 14.1 A3 B1 15.3 A4 B1 7.3
 A1 B1 11.6 A2 B1 13.8 A3 B1 15.9 A4 B1 7.8
 A1 B1 12.0 A2 B1 14.2 A3 B1 16.0 A4 B1 7.0
 A1 B2 16.5 A2 B2 19.0 A3 B2 19.5 A4 B2 8.9
 A1 B2 16.8 A2 B2 18.5 A3 B2 20.1 A4 B2 9.4
 A1 B2 16.1 A2 B2 18.2 A3 B2 19.3 A4 B2 9.3
 A1 B3 18.3 A2 B3 11.9 A3 B3 16.5 A4 B3 11.3
 A1 B3 18.7 A2 B3 12.4 A3 B3 17.2 A4 B3 10.9
 A1 B3 19.0 A2 B3 12.0 A3 B3 16.9 A4 B3 10.5

PROC GLM;

Class Plant Leaf;

MODEL Acid = Plant | Leaf(Plant);

Means Plant/Tukey;

LSMeans Leaf(Plant)/StdErr PDIF;

RUN;

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Source	DF	SS	MeanSquare	FValue	Pr>F
Model	11	530.6322222	48.2392929	381.67	<.0001
Plant	3	343.1788889	114.3929630	905.09	<.0001
Leaf(Plant)	8	187.4533333	23.4316667	185.39	<.0001
Error	24	3.0333333	0.1263889		

Tukey Grouping	Mean	N	Plant
A	17.4111	9	A3
B	15.5778	9	A1
C	14.9000	9	A2
D	9.1556	9	A4

显著. A3 >> A1 >> A2 >> A4

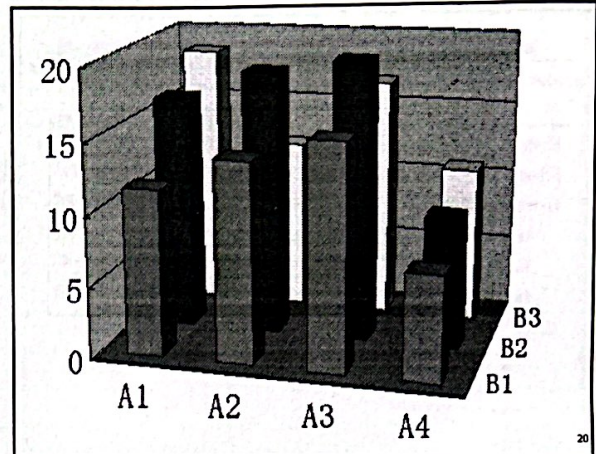
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B1	A1	11.6000000	0.2052550	<.0001	1
B2	A1	16.4666667	0.2052550	<.0001	2
B3	A1	18.6666667	0.2052550	<.0001	3
B1	A2	14.0333333	0.2052550	<.0001	4
B2	A2	18.5666667	0.2052550	<.0001	5
B3	A2	12.1000000	0.2052550	<.0001	6
B1	A3	15.7333333	0.2052550	<.0001	7
B2	A3	19.6333333	0.2052550	<.0001	8
B3	A3	16.8666667	0.2052550	<.0001	9
B1	A4	7.3666667	0.2052550	<.0001	10
B2	A4	9.2000000	0.2052550	<.0001	11
B3	A4	10.9000000	0.2052550	<.0001	12

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U	1	2	3	4	5	6	7	8	9	10	11	12
1		<.0001	<.0001	<.0001	<.0001	0.0978	<.0001	<.0001	<.0001	<.0001	<.0001	0.0239
2	<.0001		<.0001	<.0001	<.0001	<.0001	0.0185	<.0001	0.1809	<.0001	<.0001	<.0001
3	<.0001	<.0001		<.0001	0.7335	<.0001	<.0001	0.0028	<.0001	<.0001	<.0001	<.0001
4	<.0001	<.0001	<.0001		<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001
5	<.0001	<.0001	0.7335	<.0001		<.0001	<.0001	0.0012	<.0001	<.0001	<.0001	<.0001
6	0.0978	<.0001	<.0001	<.0001	<.0001		<.0001	<.0001	<.0001	<.0001	<.0001	0.0064
7	<.0001	0.0185	<.0001	<.0001	<.0001	<.0001		<.0001	0.0007	<.0001	<.0001	<.0001
8	<.0001	<.0001	0.0028	<.0001	0.0012	<.0001	<.0001		<.0001	<.0001	<.0001	<.0001
9	<.0001	0.1809	<.0001	<.0001	<.0001	<.0001	0.0007	<.0001		<.0001	<.0001	<.0001
10	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001		<.0001	<.0001
11	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001		<.0001
12	0.0239	<.0001	<.0001	<.0001	<.0001	0.0064	<.0001	<.0001	<.0001	<.0001	<.0001	

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不同植株,不同叶片均有显著差异的。

Random Model

$$Y_{ijk} = \mu + \alpha_i + \beta_{j(i)} + \epsilon_{ijk}$$

$$\sim N(\mu, \sigma_e^2 + \sigma_\beta^2 + \sigma_\alpha^2)$$

$$i = 1, 2, \dots, a; j = 1, 2, \dots, b; k = 1, 2, \dots, r$$

$$\epsilon_{ijk} \sim N(0, \sigma_e^2), \alpha_i \sim N(0, \sigma_\alpha^2), \beta_{j(i)} \sim N(0, \sigma_\beta^2)$$

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Cause of Variation	Degrees of Freedom	MS	E(MS)
A	a-1	MSA	$\sigma_e^2 + r\sigma_\beta^2 + br\sigma_\alpha^2$
B(A)	a(b-1)	MSB(A)	$\sigma_e^2 + r\sigma_\beta^2$
Error	ab(r-1)	MSE	σ_e^2

F test

$$H_0: \sigma_\alpha^2 = 0; H_1: \sigma_\alpha^2 > 0$$

$$F = \frac{MSA}{MSB(A)} \sim F(a-1, a(b-1)) \text{ if } H_0 \text{ is true}$$

$$H_0: \sigma_\beta^2 = 0; H_1: \sigma_\beta^2 > 0$$

$F = \frac{MSB(A)}{MSE} \sim F(a(b-1), ab(r-1)) \text{ if } H_0 \text{ is true}$
Note: under H_0 , the expected values of numerator and denominator in the F statistic are same

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Case Study (1)

- Training School
- Factors (random):
 - Three training schools
 - Two teachers in each school
 - 6 treatments (3 × 2)
- Study the variation for schools and teachers within school

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Data Training;
Input School \$ Inst \$ Score @@;
Datalines;

A1 B1 25 A1 B2 14
A1 B1 29 A1 B2 11
A2 B1 11 A2 B2 22
A2 B1 6 A2 B2 18
A3 B1 17 A3 B2 5
A3 B1 20 A3 B2 2

```
PROC GLM;
Class School Inst;
MODEL Score = School Inst(School);
Random School Inst(School);
Test H = School E = Inst(School);
RUN;
```

```
PROC VARCOMP method=REML;
Class School Inst;
MODEL Score = School Inst(School);
RUN;
```

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不放分, 错误.

Not Selecting for Error MS (Fixed Model)

Source	DF	Type I SS	Mean Square	F Value	Pr > F
School	2	156.5000000	78.2500000	11.18	0.0095
Inst(School)	3	547.5000000	189.1666667	27.02	0.0007

Adjusted for Error MS (Random Model)

Source	DF	Type III SS	Mean Square	F Value	Pr > F
School	2	156.5000000	78.2500000	0.41	0.6940
Inst(School)	3	587.5000000	189.1666667	27.02	0.0007

REML Iterations				
Iteration	Objective	Var(School)	Var(Inst(School))	Var(Error)
0	36.848575944	0	81.3731277533	6.2537444934
1	36.5531019312	0	69.9525133227	6.9204432456
2	36.5522284042	0	68.9116774771	6.9990958175
3	36.5522282941	0	68.8999948690	7.0000003974
4	36.5522282941	0	68.8999948690	7.0000003974

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REML Estimates

Variance Component	Estimate
Var(School)	0
Var(Inst(School))	68.89999
Var(Error)	7.00000

Asymptotic Covariance Matrix of Estimates

	Var(School)	Var(Inst(School))	Var(Error)
Var(School)	0	0	0
Var(Inst(School))	0	2100.8	-8.16666
Var(Error)	0	-8.16666	16.33334

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Case Study (2)

- ☐ Testing the acidity of plant:
- ☐ Factor (random):
 - A: plant (A1 A2 A3 A4)
 - B: leaf (B1 B2 B3)
 - 12 treatments (4 × 3)
 - 3 times repeat
- ☐ Study the variation of plants and leaves in plant

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```

Data Leaf;
Input Plant $ Leaf $ Acid @@;
Datalines;
A1 B1 11.2 A2 B1 14.1 A3 B1 15.3 A4 B1 7.3
A1 B1 11.6 A2 B1 13.8 A3 B1 15.9 A4 B1 7.8
A1 B1 12.0 A2 B1 14.2 A3 B1 16.0 A4 B1 7.0
A1 B2 16.5 A2 B2 19.0 A3 B2 19.5 A4 B2 8.9
A1 B2 16.8 A2 B2 18.5 A3 B2 20.1 A4 B2 9.4
A1 B2 16.1 A2 B2 18.2 A3 B2 19.3 A4 B2 9.3
A1 B3 18.3 A2 B3 11.9 A3 B3 16.5 A4 B3 11.3
A1 B3 18.7 A2 B3 12.4 A3 B3 17.2 A4 B3 10.9
A1 B3 19.0 A2 B3 12.0 A3 B3 16.9 A4 B3 10.5
;
PROC GLM;
Class Plant Leaf;
MODEL Acid = Plant Leaf(Plant);
Random Plant Leaf(Plant)/test;
RUN;

PROC VARCOMP Method=REML;
Class Plant Leaf;
MODEL Acid = Plant Leaf(Plant);
RUN;

```

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Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Plant	3	343.178889	114.392963	4.88	0.0324
Error	8	187.453333	23.431667		
Error:MS(Leaf(Plant))					

Source	DF	Type III	SS	Mean
Leaf(Plant)	8	187.453333	23.431667	185.39
Error:	24	3.033333	0.126389	
MS(Error)				

Iteration	Objective	Var(Plant)	(Plant)	Var(Error)
0	-10.1897751674	10.1068106996	7.7684259259	0.1263888889
1	-10.1897751674	10.1068106996	7.7684259259	0.1263888889

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REML Estimates

Variance Component	Estimate
Var(Plant)	10.10681
Var(Leaf(Plant))	7.76843
Var(Error)	0.12639

Asymptotic Covariance Matrix of Estimates

	Var(Plant)	Var(Leaf(Plant))	Var(Error)
Var(Plant)	109.39622	-5.08373	3.5052E-15
Var(Leaf(Plant))	-5.08373	15.25134	-0.0004437
Var(Error)	3.5052E-15	-0.0004437	0.0013312

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不放分, 错误.
不放分, 错误.
不放分, 错误.

Analysis by QTModel (Random Model)

QTModel

File Edit Format Run Window Help

Agency FB

```

input Plant Leaf Acid;
infile "C:/QTModel/Data/LMDB_Plant.txt" ;
proc Mixed/MIVQUE AUP GLS;
class Plant Leaf;
model Acid = Plant Leaf(Plant)/anova estimate;
random Plant Leaf(Plant)/Predict jackknife=1;

```

Plant	Leaf	Acid
1	1	11.2
1	1	11.6
1	1	12
1	2	16.5
1	2	16.8

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Variance component estimation:			
Factor	Estimate	S.E.	P-value
Plant	10.1068	0.627417	9.20821e-018
Leaf(Plant)	7.76843	0.516225	6.9253e-017
Residual	0.126389		

Random effect analysis:			
Effect	Predict	S.E.	P-value
Plant1	1.1741	0.1168	0.000000
Plant2	0.5697	0.0995	0.000002
Plant3	2.8089	0.1296	0.000000
Plant4	-4.5527	0.1424	0.000000
Leaf1*(Plant1)	-4.1788	0.2467	0.000000
Leaf1*(Plant2)	-0.8292	0.1604	0.000008
Leaf1*(Plant3)	-1.1636	0.2606	0.000068
Leaf1*(Plant4)	-3.1946	0.2978	0.000000
Leaf2*(Plant1)	1.3057	0.2365	0.000003
Leaf2*(Plant2)	4.2797	0.2410	0.000000
Leaf2*(Plant3)	3.2315	0.2812	0.000000
Leaf2*(Plant4)	-1.1285	0.2653	0.000164

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Mixed Model

$$Y_{ijk} = \mu + \alpha_i + \beta_{j(i)} + \varepsilon_{ijk}$$

$$\sim N(\mu + \alpha_i, \sigma_\varepsilon^2 + \sigma_\beta^2)$$

$$i = 1, 2, \dots, a; j = 1, 2, \dots, b; k = 1, 2, \dots, r;$$

$$\sum_i \alpha_i = 0$$

$$\varepsilon_{ijk} \sim N(0, \sigma_\varepsilon^2), \beta_{j(i)} \sim N(0, \sigma_\beta^2)$$

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Cause of Degrees of Freedom

	MS	E(MS) Mixed
A	a-1	MSA
B(A)	a(b-1)	MSB(A)
Error	ab(r-1)	MSE

F test

H_0 : all $\alpha_i = 0$; H_1 : not all $\alpha_i = 0$

$F^* = \frac{MSA}{MSB(A)} \sim F(a-1, a(b-1))$ if H_0 is true

H_0 : $\sigma_\beta^2 = 0$; H_1 : $\sigma_\beta^2 > 0$

$F^* = \frac{MSB(A)}{MSE} \sim F(a(b-1), ab(r-1))$ if H_0 is true

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Case Study (1)

□ Training school

□ Factors (mixed):

■ Fixed: three regional training schools

■ Random: two teachers in each school

■ 6 treatments (3 × 2)

□ Study the Effects of schools and variation for instructors within school

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Data Training;

Input School \$ Inst \$ Score @@;

Datalines;

A1 B1 25 A1 B2 14

A1 B1 29 A1 B2 11

A2 B1 11 A2 B2 22

A2 B1 6 A2 B2 18

A3 B1 17 A3 B2 5

A3 B1 20 A3 B2 2

;

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```

PROC GLM;
Class School Inst;
MODEL Score = School Inst(School);
Random Inst(School)/Test;
Means School/Tukey E=Inst(School);
RUN;

```

The E= option specifies the error mean square used in the multiple comparisons. By default, PROC GLM uses the overall residual or error mean square (MS). The effect specified with the E= option must be a term in the model; otherwise, the procedure uses the residual MS.

```

PROC MIXED;
Class School Inst;
MODEL Score = School; 随机化、随机效应不进去
Random Inst(School)/SOLUTION;
LSMeans School/ADJUST=TUKEY;
RUN;

```

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Proc GLM Results:

Source	Type III Expected Mean Square
School	$\text{Var}(\text{Error}) + 2 \text{Var}(\text{Inst}(\text{School})) + Q(\text{School})$
Inst(School)	$\text{Var}(\text{Error}) + 2 \text{Var}(\text{Inst}(\text{School}))$

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
School	2	156.500000	78.250000	0.41	0.6940
Error	3	567.500000	189.166667		
Error: MS(Inst(School))					
Inst(School)	3	567.500000	189.166667	27.02	0.0007
Error: MS(Error)	6	42.000000	7.000000		

Tukey	Mean	N	School
Grouping			
A	19.750	4	A1
A	14.250	4	A2
A	11.000	4	A3

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Effect	School	Inst	Estimate	SE Pred	DF	t Value	Pr > t
Inst(School)	A1	B1	6.9817	6.8722	6	1.02	0.3489
Inst(School)	A1	B2	-6.9817	6.8722	6	-1.02	0.3489
Inst(School)	A2	B1	-5.5372	6.8722	6	-0.81	0.4511
Inst(School)	A2	B2	5.5372	6.8722	6	0.81	0.4511
Inst(School)	A3	B1	7.2225	6.8722	6	1.05	0.3337
Inst(School)	A3	B2	-7.2225	6.8722	6	-1.05	0.3337

Effect	School	Estimate	SE	DF	t Value	Pr > t
School	A1	19.7500	6.8769	3	2.87	0.0639
School	A2	14.2500	6.8769	3	2.87	0.1300
School	A3	11.0000	6.8769	3	1.60	0.2080

Effect	School	School	Estimate	Error	DF	t Value	Pr > t	Adjust	Adj P
School	A1	A2	5.5000	9.7254	3	0.57	0.6112	Tukey	0.8466
School	A1	A3	8.7500	9.7254	3	0.90	0.4346	Tukey	0.6771
School	A2	A3	3.2500	9.7254	3	0.33	0.7602	Tukey	0.9415

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Case Study (2)

□ Testing plant acid level:

□ Factors (Mixed):

■ Fixed: plant (A1 A2 A3 A4)

■ Random: leaf (B1 B2 B3)

■ 12 treatments (4 × 3)

■ 3 replications

□ Study the variation of plants and leaves in plant

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```

Data Leaf;
Input Plant $ Leaf $ Acid @@;
Datalines;
A1 B1 11.2 A2 B1 14.1 A3 B1 15.3 A4 B1 7.3
A1 B1 11.6 A2 B1 13.8 A3 B1 15.9 A4 B1 7.8
A1 B1 12.0 A2 B1 14.2 A3 B1 16.0 A4 B1 7.0
A1 B2 16.5 A2 B2 19.0 A3 B2 19.5 A4 B2 8.9
A1 B2 16.8 A2 B2 18.5 A3 B2 20.1 A4 B2 9.4
A1 B2 16.1 A2 B2 18.2 A3 B2 19.3 A4 B2 9.3
A1 B3 18.3 A2 B3 11.9 A3 B3 16.5 A4 B3 11.3
A1 B3 18.7 A2 B3 12.4 A3 B3 17.2 A4 B3 10.9
A1 B3 19.0 A2 B3 12.0 A3 B3 16.9 A4 B3 10.5
;

```

```

PROC GLM;
Class Plant Leaf;
MODEL Acid = Plant Leaf(Plant);
Random Leaf(Plant)/Test;
Means Plant/Tukey E = Leaf(Plant);
RUN;
PROC MIXED;
MODEL Acid = Plant;
Random Leaf(Plant)/SOLUTION;
LSMeans Plant/ADJUST=TUKEY;
RUN;

```

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Source Type III Expected Mean Square

Source	Type III Expected Mean Square
Plant	$\text{Var}(\text{Error}) + 3 \text{Var}(\text{Leaf}(\text{Plant})) + Q(\text{Plant})$
Leaf(Plant)	$\text{Var}(\text{Error}) + 3 \text{Var}(\text{Leaf}(\text{Plant}))$

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Plant	3	343.178889	114.392963	4.88	0.0324
Error	8	187.453333	23.431667		
Error: MS(Leaf(Plant))					

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Leaf(Plant)	8	187.453333	23.431667	185.39	<.0001
Error:	24	3.033333	0.126389		
MS(Error)					

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Tukey Grouping	Mean	N	Plant
A	17.411	9	A3
B A	15.578	9	A1
B A	14.900	9	A2
B	9.156	9	A4

Cov Parm	Estimate
Leaf(Plant)	7.7684
Residual	0.1264

Estimate of the variance of random effect

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Effect	Plant	Plant	Estimate	Error	DF	t Value	Pr > t	Adjust	Adj P
Plant	A1	A2	0.6778	2.2819	8	0.30	0.7740	Tukey	0.9983
Plant	A1	A3	-1.8333	2.2819	8	-0.80	0.4449	Tukey	0.8511
Plant	A1	A4	0.4722	2.2819	8	0.21	0.8227	Tukey	0.9065
Plant	A2	A3	-2.5111	2.2819	8	-1.10	0.3031	Tukey	0.6992
Plant	A2	A4	5.7444	2.2819	8	2.52	0.0360	Tukey	0.1312
Plant	A3	A4	8.2556	2.2819	8	3.62	0.0068	Tukey	0.0280

Effect	Plant	Leaf	Estimate	SE Pred	DF	t Value	Pr > t
Leaf(Plant)	A1	B1	-3.9563	1.6178	24	-2.45	0.0223
Leaf(Plant)	A1	B2	0.8841	1.6178	24	0.55	0.5898
Leaf(Plant)	A1	B3	3.0722	1.6178	24	1.90	0.0697
Leaf(Plant)	A2	B1	-0.8620	1.6178	24	-0.53	0.5991
Leaf(Plant)	A2	B2	3.6469	1.6178	24	2.25	0.0336
Leaf(Plant)	A2	B3	-2.7849	1.6178	24	-1.72	0.0981
Leaf(Plant)	A3	B1	-1.6687	1.6178	24	-1.03	0.3126
Leaf(Plant)	A3	B2	2.2102	1.6178	24	1.37	0.1846
Leaf(Plant)	A3	B3	-0.5415	1.6178	24	-0.33	0.7408
Leaf(Plant)	A4	B1	-1.7792	1.6178	24	-1.10	0.2823
Leaf(Plant)	A4	B2	0.04420	1.6178	24	0.03	0.9784
Leaf(Plant)	A4	B3	1.7350	1.6178	24	1.07	0.2942

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Analysis by QTModel (Mixed Model)				
Means of Plant:				
Levels	Means	S.E.	P-value	
Plant1	15.5778	1.44177	4.89027e-012	
Plant2	14.9	1.442	1.45197e-011	
Plant3	17.41	1.442	2.95553e-013	
Plant4	9.156	1.442	4.54764e-007	

Pairwise comparisons among different levels of Plant:				
Critical P value adjusted by FDR at 0.05: 0.025				
i/j	1	2	3	4
1	-	0.741815	0.375505	0.003605
2	0.741815	-	0.227374	0.00835343
3	0.375505	0.227374	-	0.000318823
4	0.003605	0.00835343	0.000318823	-

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Random effect analysis:			
Effect	Predict	S.E.	P-value
Leaf1*(Plant1)	-3.7955	0.2567	0.000000
Leaf1*(Plant2)	-2.2873	0.1732	0.000000
Leaf1*(Plant3)	-0.3469	0.2528	0.000000
Leaf1*(Plant4)	-1.5006	0.2555	0.000000
Leaf2*(Plant1)	0.8482	0.2395	0.000000
Leaf2*(Plant2)	2.0384	0.2380	0.000000
Leaf2*(Plant3)	3.3744	0.2654	0.000000
Leaf2*(Plant4)	0.2487	0.2109	0.000000
Leaf3*(Plant1)	2.9474	0.2400	0.000000
Leaf3*(Plant2)	-4.1320	0.1900	0.000000
Leaf3*(Plant3)	0.7345	0.2442	0.000000
Leaf3*(Plant4)	1.8708	0.2539	0.000000

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8.2 Crossed Nested Designs

□ Experimental designs with factors not all nested are called partially nested, or crossed nested designs.

	U.S. Teams (A ₁)		Foreign Teams (A ₂)	
	Observer 1 (C ₁)	Observer 2 (C ₂)	Observer 3 (C ₃)	Observer 4 (C ₄)
Small team (B ₁)	Replication 1 Replication 2	Replication 1 Replication 2	Replication 1 Replication 2	Replication 1 Replication 2
Large team (B ₂)	Replication 1 Replication 2	Replication 1 Replication 2	Replication 1 Replication 2	Replication 1 Replication 2

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- An experiment studied the effect of cultural background on group decision making.
- Teams of students were formed and assigned a task.
- Dependent variables was the number of questions raised prior to the final group decision.

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□ Factors:

■ A: Team nationality (A1=U. S. Students, A2 = Foreign Students).

■ B: Team Size (B1 = 4 members, B2 = 8 members).

■ C: Observer (C1~C2 = U.S. Observers, C3~C4 = Foreign Observers).

■ 8 treatments (2 × 2 × 2)

□ Experimental Unit: 16 teams

□ Dependent Variable: number of questions raised prior to the final group decision

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A

	U.S. Teams (i = 1)		Foreign Teams (i = 2)	
	Observer 1 (k = 1)	Observer 2 (k = 2)	Observer 3 (k = 1)	Observer 4 (k = 2)
4 members (j = 1)	16 20	14 19	7 5	4 9
8 members (j = 2)	21 25	28 19	11 17	12 15

Factor A B C(A) A:B B:C(A)

$$Y_{ijkm} = \mu + \alpha_i + \beta_j + \gamma_{k(i)} + \alpha\beta_{ij} + \beta\gamma_{jk(i)} + \varepsilon_{ijkm}$$

$i = 1, 2, \dots, a; j = 1, 2, \dots, b; k = 1, 2, \dots, c; m = 1, 2, \dots, r$

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Partition of total sum of squares and degrees of freedom

$$\begin{aligned} \sum_i \sum_j \sum_k \sum_m (Y_{ijkm} - \bar{Y}_{....})^2 = & bcr \sum_i (\bar{Y}_{i...} - \bar{Y}_{....})^2 + acr \sum_j (\bar{Y}_{.j..} - \bar{Y}_{....})^2 + br \sum_k \sum_i (\bar{Y}_{i.k.} - \bar{Y}_{i...})^2 \\ & + cr \sum_j \sum_i (\bar{Y}_{i.j.} - \bar{Y}_{i...} - \bar{Y}_{.j..} + \bar{Y}_{....})^2 + r \sum_k \sum_j \sum_i (\bar{Y}_{i.j.k} - \bar{Y}_{i.j.} - \bar{Y}_{.j.k.} + \bar{Y}_{i.j..} + \bar{Y}_{.j.k.} - \bar{Y}_{....})^2 \\ & + r \sum_k \sum_j \sum_i \sum_m (Y_{ijkm} - \bar{Y}_{ijk.})^2 \end{aligned}$$

$$SSTO = SSA + SSB + SSC(A) + SSAB + SSBC(A) + SSE$$

$$\begin{aligned} df_{TO} &= df_A + df_B + df_{C(A)} + df_{AB} + df_{BC(A)} + df_e \\ abr - 1 &= (a-1) + (b-1) + a(c-1) + (a-1)(b-1) \\ &\quad + a(b-1)(c-1) + abc(r-1) \end{aligned}$$

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Anova table for crossed nested design (fixed model)

Source	df	E(MS) Fixed
A	a-1	$\sigma_e^2 + \frac{bcr}{a-1} \sum_i \alpha_i^2$
B	b-1	$\sigma_e^2 + \frac{acr}{b-1} \sum_j \beta_j^2$
C(A)	a(c-1)	$\sigma_e^2 + \frac{br}{a(c-1)} \sum_i \sum_k \gamma_{ik(i)}^2$
AB	(a-1)(b-1)	$\sigma_e^2 + \frac{cr}{(a-1)(b-1)} \sum_i \sum_j \alpha\beta_{ij}^2$
BC(A)	a(b-1)(c-1)	$\sigma_e^2 + \frac{r}{a(b-1)(c-1)} \sum_i \sum_j \sum_k \beta\gamma_{jk(i)}^2$
Error	abc(r-1)	σ_e^2

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Source	df	E(MS) Fixed
A	a-1	$\sigma_e^2 + \frac{bcr}{a-1} \sum_i \alpha_i^2$
B	b-1	$\sigma_e^2 + \frac{acr}{b-1} \sum_j \beta_j^2$
C(A)	a(c-1)	$\sigma_e^2 + \frac{br}{a(c-1)} \sum_i \sum_k \gamma_{ik(i)}^2$
AB	(a-1)(b-1)	$\sigma_e^2 + \frac{cr}{(a-1)(b-1)} \sum_i \sum_j \alpha\beta_{ij}^2$
BC(A)	a(b-1)(c-1)	$\sigma_e^2 + \frac{r}{a(b-1)(c-1)} \sum_i \sum_j \sum_k \beta\gamma_{jk(i)}^2$
Error	abc(r-1)	σ_e^2

F test

H_0 : all $\alpha_i = 0$; H_1 : not all $\alpha_i = 0$

$F = MSA/MSE \sim F(a-1, abc(r-1))$ if H_0 is true

H_0 : all $\beta_j = 0$; H_1 : not all $\beta_j = 0$

$F = MSB/MSE \sim F(b-1, abc(r-1))$ if H_0 is true

H_0 : all $\gamma_{ik(i)} = 0$; H_1 : not all $\gamma_{ik(i)} = 0$

$F = MSC(A)/MSE \sim F(a(c-1), abc(r-1))$ if H_0 is true

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Source	df	E(MS) Fixed
A	a-1	$\sigma_e^2 + \frac{bcr}{a-1} \sum_i \alpha_i^2$
B	b-1	$\sigma_e^2 + \frac{acr}{b-1} \sum_j \beta_j^2$
C(A)	a(c-1)	$\sigma_e^2 + \frac{br}{a(c-1)} \sum_i \sum_k \gamma_{ik(i)}^2$
AB	(a-1)(b-1)	$\sigma_e^2 + \frac{cr}{(a-1)(b-1)} \sum_i \sum_j \alpha\beta_{ij}^2$
BC(A)	a(b-1)(c-1)	$\sigma_e^2 + \frac{r}{a(b-1)(c-1)} \sum_i \sum_j \sum_k \beta\gamma_{jk(i)}^2$
Error	abc(r-1)	σ_e^2

F test

H_0 : all $\alpha\beta_{ij} = 0$; H_1 : not all $\alpha\beta_{ij} = 0$

$F = MSAB/MSE \sim F((a-1)(b-1), abc(r-1))$ if H_0 is true

H_0 : all $\beta\gamma_{jk} = 0$; H_1 : not all $\beta\gamma_{jk} = 0$

$F = MSBC(A)/MSE \sim F(a(b-1)(c-1), abc(r-1))$ if H_0 is true

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Random Model

Source	E(MS) Random
A	$\sigma_e^2 + r\sigma_{\beta\gamma}^2 + cr\sigma_{\alpha\beta}^2 + br\sigma_{\gamma}^2 + bcr\sigma_{\alpha}^2$
B	$\sigma_e^2 + r\sigma_{\beta\gamma}^2 + cr\sigma_{\alpha\beta}^2 + acr\sigma_{\beta}^2$
C(A)	$\sigma_e^2 + r\sigma_{\beta\gamma}^2 + br\sigma_{\gamma}^2$
AB	$\sigma_e^2 + r\sigma_{\beta\gamma}^2 + cr\sigma_{\alpha\beta}^2$
BC(A)	$\sigma_e^2 + r\sigma_{\beta\gamma}^2$
Error	σ_e^2

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合并平方

F test

$H_0: \sigma^2 = 0; H_1: \sigma^2 > 0$

$F^* = MSA/MSCI \sim F(a-1, df_{CI})$ if H_0 is true

$$MSCI = MSC(A) + MSAB - MSBC(A)$$

$$df_{CI} = \frac{(MSC(A) + MSAB - MSBC(A))^2}{\frac{(MSC(A))^2}{df_{C(A)}} + \frac{(MSAB)^2}{df_{AB}} + \frac{(MSBC(A))^2}{df_{BC(A)}}}$$

$H_0: \sigma_{\beta}^2 = 0; H_1: \sigma_{\beta}^2 > 0$

$F^* = MSB/MSAB \sim F(b-1, (a-1)(b-1))$ if H_0 is true

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$H_0: \sigma_{\gamma}^2 = 0; H_1: \sigma_{\gamma}^2 > 0$

$F^* = MSC(A)/MSBC(A) \sim F(a(c-1), a(b-1)(c-1))$ if H_0 is true

$H_0: \sigma_{\alpha\beta}^2 = 0; H_1: \sigma_{\alpha\beta}^2 > 0$

$F^* = MSAB/MSBC(A) \sim F((a-1)(b-1), a(b-1)(c-1))$ if H_0 is true

$H_0: \sigma_{\beta\gamma}^2 = 0; H_1: \sigma_{\beta\gamma}^2 > 0$

$F^* = MSBC(A)/MSE \sim F(a(b-1)(c-1), abc(r-1))$ if H_0 is true

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Experiment Over Locations or Years

□ For studies of the effects of treatments over locations or years.

□ Factors:

■ Location/Year; Block within location/year; Treatment

Location 1			Location 2		
B1(L1)	B2(L1)	B3(L1)	B1(L2)	B2(L2)	B3(L2)
TRT 2	TRT 6	TRT 1	TRT 6	TRT 4	TRT 3
TRT 3	TRT 2	TRT 4	TRT 4		TRT 6
TRT 1	TRT 1	TRT 6	TRT 3	TRT 6	TRT 4
TRT 5	TRT 4	TRT 3	TRT 2	TRT 2	TRT 2
TRT 4	TRT 5	TRT 2	TRT 5	TRT 1	TRT 5
TRT 6	TRT 3	TRT 5	TRT 1	TRT 5	TRT 1

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fixed model

Location Block within Location Variety Interaction between Variety and Location

$$Y_{ijk} = \mu + \alpha_i + \rho_{j(l)} + \tau_k + \alpha\tau_{ik} + \varepsilon_{ijk}$$

$i = 1, 2, \dots, l; j = 1, 2, \dots, b$
 $k = 1, 2, \dots, t$

$\alpha_i \sim N(0, \sigma_{\alpha}^2), \rho_{j(l)} \sim N(0, \sigma_{\rho}^2)$
 $\varepsilon_{ijk} \sim N(0, \sigma_{\varepsilon}^2)$
 $\sum_k \tau_k = 0$

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Partition of total sum of squares and degrees of freedom

$$\sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{...})^2 =$$

$$bl \sum_i (\bar{Y}_{i..} - \bar{Y}_{...})^2 + t \sum_j (\bar{Y}_{.j.} - \bar{Y}_{...})^2 + bl \sum_k (\bar{Y}_{..k} - \bar{Y}_{...})^2$$

$$+ b \sum_i \sum_k (\bar{Y}_{i.k} - \bar{Y}_{i..} - \bar{Y}_{..k} + \bar{Y}_{...})^2$$

$$+ \sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{..k} + \bar{Y}_{...})^2$$

$$SSTO = SSL + SSB(L) + SST + SSLT + SSE$$

$$df_{TO} = df_L + df_{B(L)} + df_T + df_{LT} + df_{\varepsilon}$$

$$lbt - 1 = (l - 1) + l(b - 1) + (t - 1) + (l - 1)(t - 1) + l(b - 1)(t - 1)$$

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Experiment Over Locations and Years

For studies of the effects of treatments over locations and years.

□ Factors:

- Location
- Year
- Block within location and year
- Treatment

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Year 1			Year 2		
Location 1			Location 2		
B1(Y1L1)	B2(Y1L1)	B3(Y1L1)	B1(Y1L2)	B2(Y1L2)	B3(Y1L2)
TRT 1	TRT 6	TRT 1	TRT 6	TRT 4	TRT 3
TRT 3	TRT 2	TRT 4	TRT 4	TRT 3	TRT 6
TRT 1	TRT 1	TRT 6	TRT 3	TRT 6	TRT 4
TRT 5	TRT 4	TRT 3	TRT 2	TRT 2	TRT 2
TRT 4	TRT 5	TRT 2	TRT 5	TRT 1	TRT 5
TRT 6	TRT 3	TRT 5	TRT 1	TRT 5	TRT 1

Year 2			Year 1		
Location 1			Location 2		
B1(Y2L1)	B2(Y2L1)	B3(Y2L1)	B1(Y2L2)	B2(Y2L2)	B3(Y2L2)
TRT 1	TRT 1	TRT 6	TRT 3	TRT 6	TRT 4
TRT 5	TRT 4	TRT 3	TRT 2	TRT 2	TRT 2
TRT 2	TRT 6	TRT 1	TRT 6	TRT 4	TRT 3
TRT 6	TRT 3	TRT 5	TRT 1	TRT 5	TRT 1
TRT 3	TRT 2	TRT 4	TRT 4	TRT 3	TRT 6
TRT 4	TRT 5	TRT 2	TRT 5	TRT 1	TRT 5

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fixed effect

Variety Location Year

$$Y_{hijk} = \mu + \tau_h + \alpha_i + \beta_j + \tau\alpha_{hi} + \tau\beta_{hj} + \alpha\beta_{ij} + \tau\alpha\beta_{hij} + \rho_{k(ij)} + \varepsilon_{hijk}$$

Block within Location and year

$h = 1, 2, \dots, t; i = 1, 2, \dots, l; j = 1, 2, \dots, y; k = 1, 2, \dots, b$

$\alpha_i \sim N(0, \sigma_\alpha^2), \beta_j \sim N(0, \sigma_\beta^2)$

$\rho_{k(ij)} \sim N(0, \sigma_\rho^2), \varepsilon_{hijk} \sim N(0, \sigma_\varepsilon^2)$

$\sum_h \tau_h = 0$

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ANOVA Table for Regional Trial

Source	df	MS	EMS
Genotype	$g-1$	MSG	$\sigma_\varepsilon^2 + b\sigma_{\alpha\tau}^2 + lb\sigma_{\alpha\tau}^2 + yb\sigma_{\alpha\tau}^2 + \frac{y}{g-1} \sum G_i^2$
Year	$y-1$	MSY	$\sigma_\varepsilon^2 + g\sigma_{\beta\tau}^2 + gb\sigma_{\beta\tau}^2 + \frac{g}{y-1} \sum Y_j^2$
Location	$l-1$	MSL	$\sigma_\varepsilon^2 + g\sigma_{\alpha\tau}^2 + gb\sigma_{\alpha\tau}^2 + \frac{g}{l-1} \sum L_i^2$
Y×L	$(y-1)(l-1)$	MSYL	$\sigma_\varepsilon^2 + g\sigma_{\beta\tau}^2 + gb\sigma_{\beta\tau}^2$
G×Y	$(g-1)(y-1)$	MSGY	$\sigma_\varepsilon^2 + b\sigma_{\alpha\tau}^2 + lb\sigma_{\alpha\tau}^2$
G×L	$(g-1)(l-1)$	MSGL	$\sigma_\varepsilon^2 + b\sigma_{\alpha\tau}^2 + yb\sigma_{\alpha\tau}^2$
G×Y×L	$(g-1)(y-1)(l-1)$	MSGYL	$\sigma_\varepsilon^2 + b\sigma_{\alpha\tau}^2$
Block	$y(l-1)$	MSB(VL)	$\sigma_\varepsilon^2 + g\sigma_{\rho}^2$
Error	$yl(g-1)(b-1)$	MSe	σ_ε^2

$$F = \frac{MSG}{MSe} \Rightarrow \frac{\sigma_\varepsilon^2 + b\sigma_{\alpha\tau}^2 + lb\sigma_{\alpha\tau}^2 + yb\sigma_{\alpha\tau}^2 + \frac{y}{g-1} \sum G_i^2}{\sigma_\varepsilon^2}$$

$$F = \frac{MSG}{MSC} \Rightarrow \frac{\sigma_\varepsilon^2 + b\sigma_{\alpha\tau}^2 + lb\sigma_{\alpha\tau}^2 + yb\sigma_{\alpha\tau}^2 + \frac{y}{g-1} \sum G_i^2}{\sigma_\varepsilon^2 + b\sigma_{\alpha\tau}^2 + lb\sigma_{\alpha\tau}^2 + yb\sigma_{\alpha\tau}^2}$$

$$F = \frac{MSG}{MSGY + MSGL - MSGYL} \quad df_1 = g-1$$

$$df_2 = \frac{(MSGY)^2 + (MSGL)^2 - (MSGYL)^2}{4gy} + \frac{(MSGY)^2 + (MSGL)^2 - (MSGYL)^2}{4gl} + \frac{(MSGY)^2 + (MSGL)^2 - (MSGYL)^2}{4gyl}$$

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How to make statistical inference for selecting the superior genotypes in regional trial test (mixed model)

Genotype	GXY	GXL	GXY×L	Statistical Inference
NS	NS	NS	*	Select superior genotypes in specific location and year
NS	NS	*	*/NS	Select superior genotypes in specific location
-	*	*/NS	*/NS	Select superior genotypes in specific most of years
*	*/NS	*/NS	*/NS	Select superior genotypes

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□ A study of the effects of variety, year, and location on rice yield.

□ Factors:

- 5 varieties (fixed)
- 2 years (random)
- 4 locations (random)
- 3 blocks (random)

□ 40 treatments (5 × 2 × 4)

□ Experimental Unit: 120 plots

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实验资料: Variety Year Location Block Yield

```

1 1 1 1 19.7 1 2 3 3 30.8 2 2 2 2 61.1
1 1 1 2 31.4 1 2 4 1 26.6 2 2 2 3 52.2
1 1 1 3 29.6 1 2 4 2 26.5 2 2 3 1 38.3
1 1 2 1 40.8 1 2 4 3 32.7 2 2 3 2 29.4
1 1 2 2 29.4 2 1 1 1 28.6 2 2 3 3 30.5
1 1 2 3 30.2 2 1 1 2 38.3 2 2 4 1 21.4
1 1 3 1 34.7 2 1 1 3 43.5 2 2 4 2 18.7
1 1 3 2 29.1 2 1 2 1 44.4 2 2 4 3 24.1
1 1 3 3 35.1 2 1 2 2 34.9 3 1 1 1 20.3
1 1 4 1 20.2 2 1 2 3 33.9 3 1 1 2 27.5
1 1 4 2 30.2 2 1 3 1 28.8 3 1 1 3 32.6
1 1 4 3 16 2 1 3 2 28.7 3 1 2 1 44.6
1 2 1 1 45.5 2 1 3 3 21 3 1 2 2 41.4
1 2 1 2 50.3 2 1 4 1 13.2 3 1 2 3 26.2
1 2 1 3 60 2 1 4 2 20.5 3 1 3 1 29.8
1 2 2 1 53.9 2 1 4 3 9.6 3 1 3 2 38.4
1 2 2 2 58.8 2 2 1 1 47.5 3 1 3 3 28
1 2 2 3 47.7 2 2 1 2 41.1 3 1 4 1 24.5
1 2 3 1 42.1 2 2 1 3 49.4 3 1 4 2 41.6
1 2 3 2 47.1 2 2 2 1 63.7 3 1 4 3 30.6

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3 2 1 1 54.2 4 1 3 3 20.4 5 1 2 2 47.6
3 2 1 2 52.3 4 1 4 1 19 5 1 2 3 55.4
3 2 1 3 64.5 4 1 4 2 18.4 5 1 3 1 43
3 2 2 1 53.9 4 1 4 3 24.6 5 1 3 2 32.7
3 2 2 2 59.1 4 2 1 1 62.2 5 1 3 3 32
3 2 2 3 56.4 4 2 1 2 53.1 5 1 4 1 27.6
3 2 3 1 42.1 4 2 1 3 74.7 5 1 4 2 30
3 2 3 2 40 4 2 2 1 74.2 5 1 4 3 22.7
3 2 3 3 39.8 4 2 2 2 75.6 5 2 1 1 47.4
3 2 4 1 20.7 4 2 2 3 67 5 2 1 2 57.8
3 2 4 2 26.8 4 2 3 1 44.3 5 2 1 3 50.5
3 2 4 3 30.4 4 2 3 2 43.5 5 2 2 1 51.1
4 1 1 1 27.9 4 2 3 3 47.7 5 2 2 2 47.3
4 1 1 2 40 4 2 4 1 20.7 5 2 2 3 45
4 1 1 3 46.1 4 2 4 2 23.6 5 2 3 1 53.9
4 1 2 1 39.8 4 2 4 3 30.9 5 2 3 2 51.8
4 1 2 2 39.2 5 1 1 1 22.3 5 2 3 3 50.3
4 1 2 3 29.1 5 1 1 2 30.8 5 2 4 1 32.6
4 1 3 1 27.2 5 1 1 3 31.1 5 2 4 2 40
4 1 3 2 27.6 5 1 2 1 71.5 5 2 4 3 34.2

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PROC GLM;
Class Gen Year Loc Block;
MODEL Yield = Gen|Year|Loc Block(Year*Loc);
Random Year Loc Year*Loc Gen*Year Gen*Loc
Gen*Year*Loc Block(Year*Loc)/test;
Means Gen/Tukey;
RUN;
PROC Mixed;
Class Gen Year Loc Block;
MODEL Yield = Gen;
Random Year Loc Year*Loc Gen*Year Gen*Loc
Gen*Year*Loc Block(Year*Loc)/SOLUTION;
LsMeans Gen/ADJUST=Tukey;
RUN;
PROC VARCOMP method=REML;
Class Gen Year Loc Block;
MODEL Yield = Gen|Year|Loc
Block(Year*Loc)/fixed = 1;
RUN;

```

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SAS Test

Source	DF	Fixed Model		Mixed Model	
		F Value	P-value	F Value	P-value
Gen	4	12.71	<.0001	3.48	0.480
Gen*Year	4	6.82	0.0001	0.77	0.565
Gen*Loc	12	5.69	<.0001	0.64	0.7724
Gen*Year*Loc	12	8.85	<.000	8.85	<.000

SAS Incorrect Test

Tukey Grouping	Mean	N	Gen
A	42.025	24	5
A	40.700	24	4
B A	38.571	24	3
B C	36.183	24	1
C	34.283	24	2

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Results of Proc Mixed

The Mixed Procedure

Covariance Parameter Estimates

Cov Parm	Estimate
Year	89.8686
Location	87.7911
Year*Location	16.7567
Variety*Year	0
Variety*Location	0
Variety*Year*Location	39.3758
Block(Year*Location)	17.4660
Residual	19.8295

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Results of Proc Mixed

Effect	Variety	Year	Location	Block	Estimate	Prod	DF	t Value	Pr > t
Variety*Year*Location	1	1	1		-3.8013	3.8434	04	-0.78	0.4377
Variety*Year*Location	1	1	2		-4.3621	3.8434	04	-1.14	0.2585
Variety*Year*Location	1	1	3		4.1895	3.8434	04	1.07	0.2890
Variety*Year*Location	1	1	4		1.4941	3.8434	04	0.39	0.6988
Variety*Year*Location	1	2	1		1.4583	3.8434	04	0.38	0.7060
Variety*Year*Location	1	2	2		-0.3062	3.8434	04	-0.23	0.8156
Variety*Year*Location	1	2	3		-0.1638	3.8434	04	-0.04	0.9681
Variety*Year*Location	1	2	4		1.2875	3.8434	04	0.36	0.7193
Variety*Year*Location	2	1	1		7.1616	3.8434	04	1.86	0.0670
Variety*Year*Location	2	1	2		0.3298	3.8434	04	0.24	0.8090
Variety*Year*Location	2	1	3		-0.1107	3.8434	04	-0.03	0.9771
Variety*Year*Location	2	1	4		-3.5812	3.8434	04	-0.91	0.3657
Variety*Year*Location	2	2	1		-2.0174	3.8434	04	-0.52	0.6015
Variety*Year*Location	2	2	2		3.5818	3.8434	04	1.93	0.1571
Variety*Year*Location	2	2	3		-4.7853	3.8434	04	-1.25	0.2178
Variety*Year*Location	2	2	4		-3.1772	3.8434	04	-0.83	0.4118
Variety*Year*Location	3	1	1		-5.1437	3.8434	04	-1.34	0.1855
Variety*Year*Location	3	1	2		-3.6587	3.8434	04	-0.73	0.4683
Variety*Year*Location	3	1	3		1.2781	3.8434	04	0.33	0.7406
Variety*Year*Location	3	1	4		8.1363	3.8434	04	2.12	0.0382

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SAS REML Estimation

Variance Component	Estimate
Var(Year)	89.06860
Var(Loc)	87.79112
Var(Year*Loc)	16.75673
Var(Gen*Year)	0
Var(Gen*Loc)	0
Var(Gen*Year*Loc)	39.37586
Var(Block(Year*Loc))	17.46600
Var(Error)	19.02950

not significant for all variances estimation

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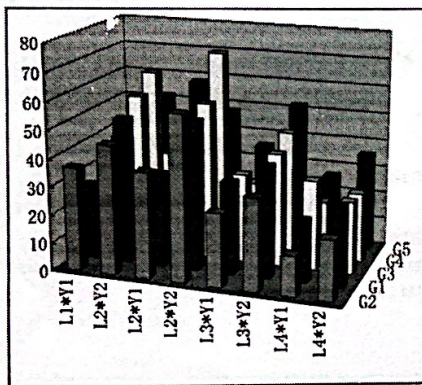
SAS REML Estimation

Asymptotic Covariance Matrix of Estimates

	Var(Year*Location)	Var(Variety*Year*Location)
Var(Year)	-167.71991	-5.0423E-8
Var(Variety*Year)	0	0
Var(Location)	-335.43982	-4.97E-8
Var(Variety*Location)	0	0
Var(Year*Location)	683.18665	-30.11189
Var(Variety*Year*Location)	-30.11189	150.55945
Var(Block(Year*Location))	-19.60479	0.75442
Var(Error)	0.75442	-3.77218

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- 针对多年多点的品种区域试验资料，品种效应为固定效应，其它效应为随机效应
- 写出分析的模型
- 写出各项效应的期望均方
- 写出SAS软件给出的期望均方表，并对各项期望均方的正确与否进行评述

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